

EARTH'S STRUCTURE

Our planet has a diverse surface, with a huge variety of landscapes and materials, ranging from water and gases, to living matter, soil, ice, and rock. However, its interior is much less varied, consisting mainly of just rock, metal, and some water.

Internal structure

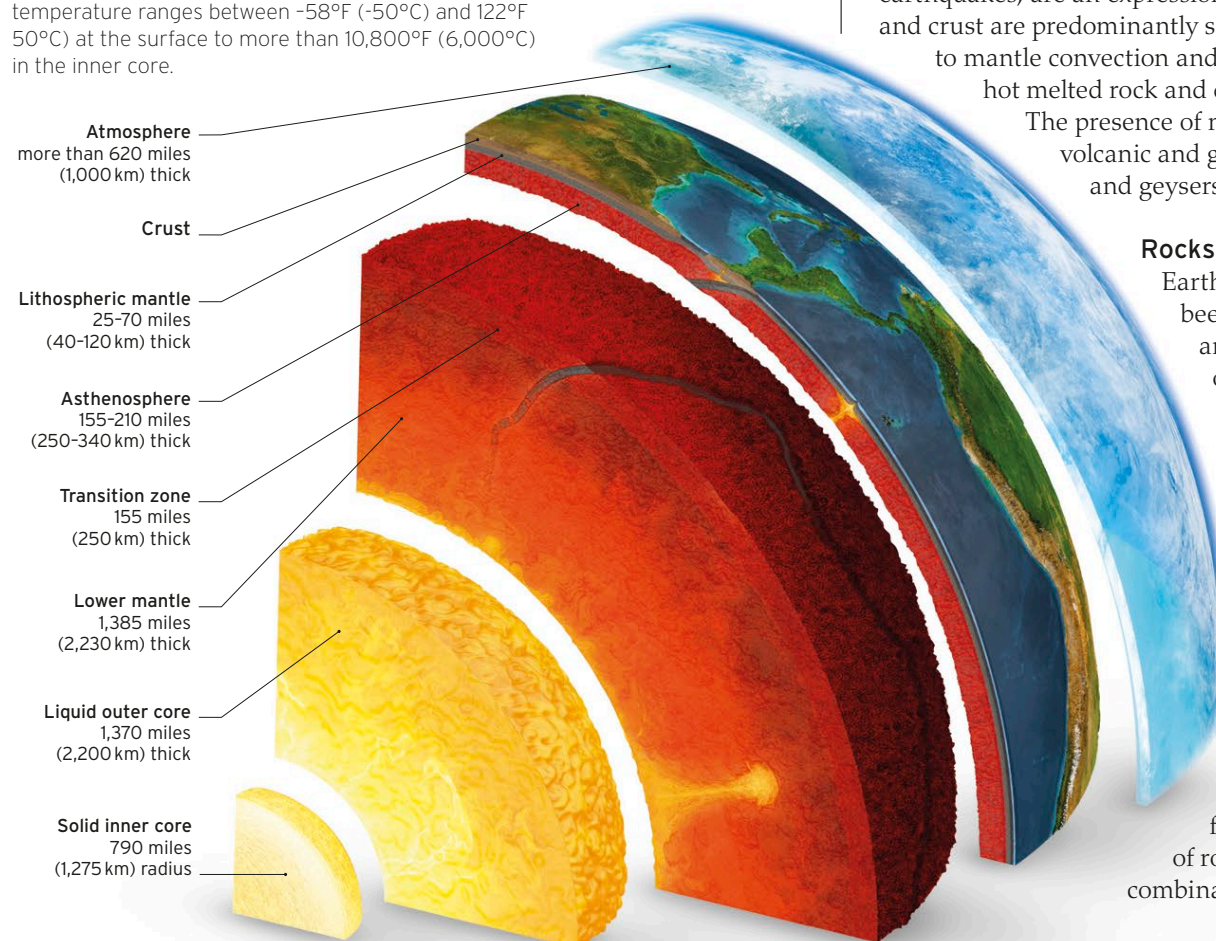
Much of what is known about Earth's structure has been established from the study of seismic waves created by earthquakes. Internally, our planet has three main layers: the core, mantle, and crust. The core is made mainly of iron, with some nickel, and is itself double-layered—a liquid outer core enclosing a solid inner core.

Surrounding the core is the mantle, which consists mainly of solid (but in parts, deformable) silicate rock. Its top part is called the upper mantle and has two layers. The uppermost of these, sometimes called lithospheric mantle, is solid and brittle and is fused to the crust. Below it is a more deformable layer, the asthenosphere. Below the upper mantle is a region called the transition zone, now known to contain significant amounts of water “locked into” its rocks. Beneath this is the thickest of all of Earth's internal layers, the lower mantle.

The outermost layer is the crust. This is solid, and there are two different kinds. Thick continental crust, made up of many different rock types, forms the land surface, while thin, relatively dense oceanic crust—containing just a few rock types—lies under the oceans. Both types are fused to the underlying lithospheric mantle, forming a combined rigid shell, the lithosphere.

▽ EARTH'S LAYERED STRUCTURE

Each layer from the crust inward has a temperature and density higher than the layer surrounding it. The temperature ranges between -58°F (-50°C) and 122°F (50°C) at the surface to more than $10,800^{\circ}\text{F}$ ($6,000^{\circ}\text{C}$) in the inner core.

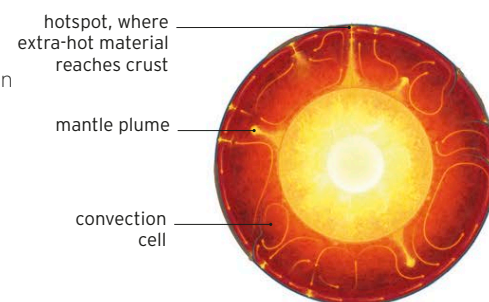


Mantle convection

A slow, gradual circulation of rock occurs in the mantle. This circulation is called mantle convection and is driven by heat flowing out of Earth's core. The mantle is also thought to contain plumes of rising, hotter, semisolid or liquid rock, called mantle plumes. Where they penetrate into the crust, these create surface hotspots (see p.13). Earth holds a colossal amount of internal heat energy, much of it generated during the planet's original formation and still trapped inside. This energy is continually added to by the radioactive decay of unstable isotopes of various chemical elements scattered throughout

▷ WHOLE-EARTH CONVECTION

The overall pattern of convection in Earth's mantle is not fully understood. One hypothesis, shown here, suggests that a series of convection cells slowly carry rock from the bottom of the mantle to the top and then back again.



the interior. Earth's energy is constantly trying to escape, and mantle convection and mantle plumes, as well as phenomena such as earthquakes, are an expression of this. Although our planet's mantle and crust are predominantly solid, as a result of processes connected to mantle convection and plumes both contain collections of hot melted rock and dissolved gases, called magma.

The presence of magma in the crust is associated with volcanic and geothermal activity (such as hot springs and geysers) at Earth's surface.

Rocks and the rock cycle

Earth's crust, the part of our planet that has been shaped into a myriad of landscapes and physical features, is made up mostly of rocks. Rocks are assemblages of chemical substances called minerals, and are of three main types: igneous, sedimentary, and metamorphic. Igneous rocks are the result of magma from deep within the planet being erupted at the surface or injected into the crust and then cooling and solidifying. Sedimentary rocks are formed by the deposition and cementation of mineral or rock particles created by the weathering or breakdown of other rocks. Metamorphic rocks are formed by the alteration of other types of rocks by high temperature, pressure, or a combination of the two. A variety of processes